

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project — Week 5 React Native (VKU Room Booking)
Team / Student Name: Nguyễn Minh Duy
Submission Date: 17/09/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

Team Members:

1. Nguyễn Minh Duy — Student ID: 23IT038 — Role: Solo (UI, Expo, booking rules, Cloudflare KV) — Contribution: 100%

Live Demo URL: <https://vku-room-booking.pages.dev/>

GitHub Repository: <https://github.com/minhduy6868/mob-lab-room-booking>

Video Demo: Not submitted.

Demo login (pre-filled): email `abc@vku.udn.vn` — password `vku@2026` — profile Nguyễn Minh Duy / 23IT038.

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Expo / React Native app	Complete	SDK 57, RN 0.86, TypeScript. Stack + Tabs (React Navigation 7).
2	Room list + search/filter	Complete	22 rooms, chips (type/building/status), Safe Area, Ionicons.
3	Booking by time slot	Complete	Today/tomorrow, ca 1–5, purpose, booking code + QR pass.
4	Overlap / conflict rules	Complete	Occupied slot, cross-room overlap, 2 slots/day, past slot, maintenance, cancel 30 min, check-in 10 min. Client + KV Functions.
5	Auth + profile	Complete	Email <code>@vku.udn.vn</code> . Google button hidden. Zustand persist session.
6	Cloud persistence	Complete	Cloudflare Pages Functions + KV VKU_BOOKINGS. GET/POST/PUT /api/bookings.
7	HTTPS live deploy	Complete	https://vku-room-booking.pages.dev/ Health: /api/health
8	Realtime occupancy	Complete	Poll KV every 2.5s after login so two clients share the same holds.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

Client is Expo (React Native Web on Pages, same codebase for Expo Go). `useAuthStore` and `useBookingStore` (Zustand) call Cloudflare. `evaluateBooking` runs on the device and again in `functions/_lib/rules.js` so a second browser cannot double-book.

Directory map:

App.tsx Font boot (Ionicons) + splash + NavigationContainer
src/screens/ Login, BrowseRooms, MyBookings, Profile
src/store/useAuthStore.ts Session persist (SecureStore / localStorage)
src/store/useBookingStore.ts hydrate / book / cancel / check-in
src/lib/booking-rules.ts Conflict engine (client)
functions/api/bookings.js GET list, POST create, PUT cancel|checkin
functions/api/auth.js Email demo `abc@vku.udn.vn`
functions/_lib/store.js KV prefix + seed Minh & Huy
scripts/prepare-pages.cjs bundle/ + icon-fonts for Pages
wrangler.toml KV binding VKU_BOOKINGS

State flow: login -> `hydrateFromCloud()` -> `decorateSlots` from KV -> POST /api/bookings -> 200 + `bookingCode` or 409 + Vietnamese message -> `refetch`. Occupancy is not local-only.

Exception handling: Functions return JSON `{ ok:false, error }`. The UI shows the error on the booking sheet. A

write is accepted only after KV putBooking. Seed rows (Minh ca 2 Lab A3-101, Huy ca 1 Library) stay as collision fixtures.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

Live host <https://vku-room-booking.pages.dev/> on a 390x844 viewport. Captured 17/09/2026 if Chromium was available.

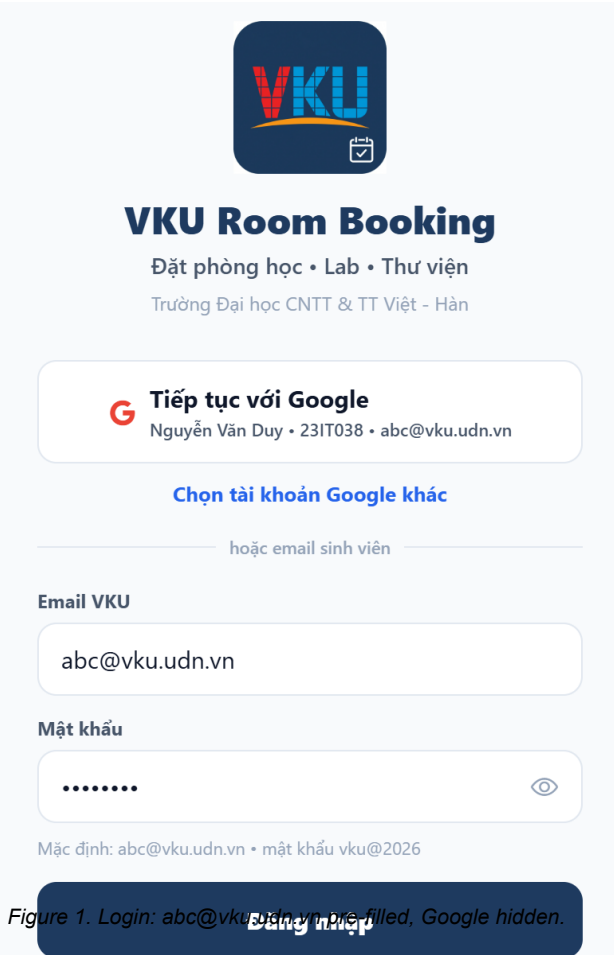


Figure 1. Login: abc@vku.udn.vn pre-filled, Google hidden.

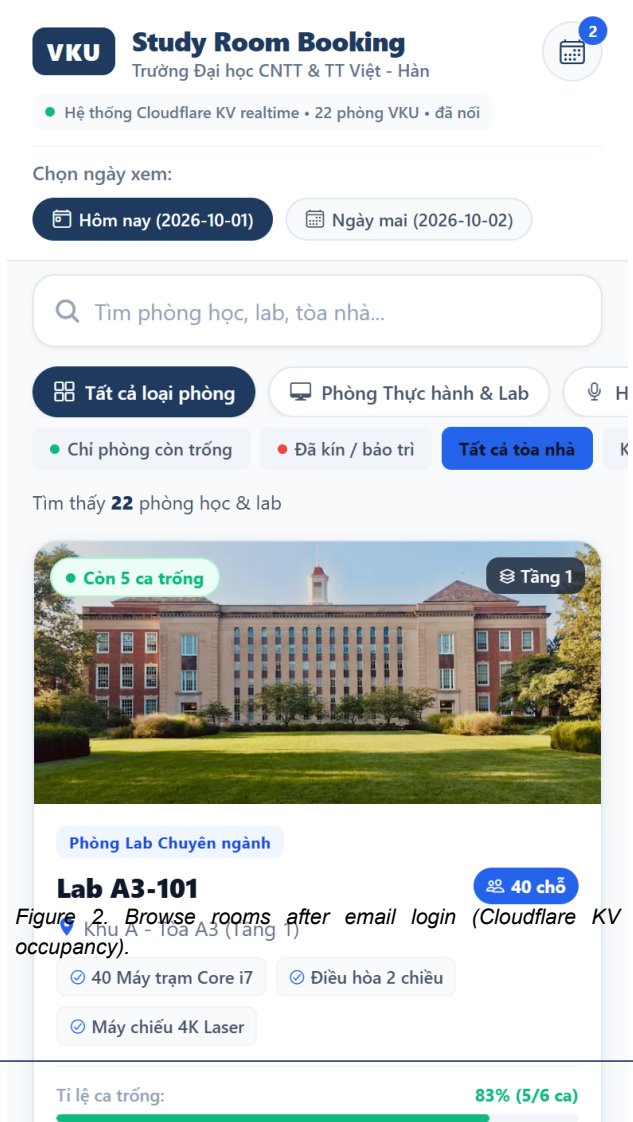


Figure 2. Browse rooms after email login (Cloudflare KV occupancy).

5. TECHNICAL CHALLENGES & RESOLUTIONS

5.1 Icons missing on Cloudflare Pages

Expo web emits `_expo/static/js` and fonts under `node_modules/@expo/.../*.ttf`. Pages reserved `*` paths and SPA rewrite `/* -> index.html` served HTML for JS/TTF, so every icon broke. Change: `scripts/prepage-pages.cjs` renames `_expo` to `/bundle`, copies hashed fonts to `/assets/icon-fonts/`, and does not install a catch-all 200 rewrite.

5.2 Logout and booking confirm did nothing on web

React Native `Alert.alert` on web is `window.alert`; button `onPress` is ignored. Change: `src/lib/confirm.ts` uses `window.confirm` on `EXPO_OS === web`.

5.3 POST /api/bookings 409 past-slot at night

Correct rule: `Asia/Ho_Chi_Minh`, a finished ca cannot be booked. Use the Tomorrow date chip to demo a live reservation after hours.